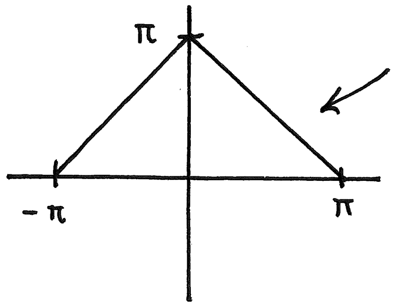
HW 14 Solutions

Problem 1

(i)



$\begin{cases} \text{Even} \\ L = \pi \end{cases}$

$$d_0 = \frac{\pi}{2}$$

$$d_n = \frac{1}{\pi}\int_0^{\pi} (\pi - x)\cos(nx)\,dx$$

$$= \text{(Integration by Parts)}$$

w/ $u = \pi - x$ $\quad dv = \cos(nx)dx$

$$= \frac{1}{\pi n^2}\left(1-(-1)^n\right) = \begin{cases} 0 & n \text{ even} \\ \frac{2}{\pi n^2} & n \text{ odd} \end{cases}$$

or

$$\circledast \quad d_n = \begin{cases} \frac{\pi}{2} & n = 0 \\ \frac{1}{\pi n^2}\left(1-(-1)^n\right) & n \neq 0 \end{cases}$$

Acceptable Answers :

1. $f(x) = \frac{\pi}{2} + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} d_n e^{inx}$

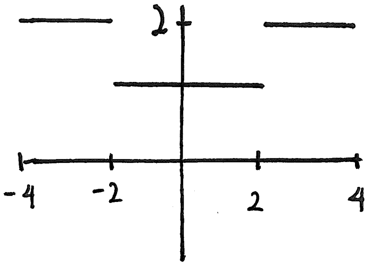
2. $f(x) = \frac{\pi}{2} + \sum_{\pm \text{odd}} \frac{2}{\pi n^2} e^{inx}$

3. $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{inx}$ (Only use this if you have also written $\circledast$)

4. $f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} 2 d_n \cos(nx)$

5. $f(x) = \frac{\pi}{2} + \sum_{+\text{odd}} \frac{4}{\pi n^2} \cos(nx)$

(ii)



$\leftarrow \begin{cases} \text{Even} \\ L = 4 \end{cases}$

$$d_0 = \frac{3}{2}$$

$$d_n = \frac{1}{4}\left[\int_0^2 \cos\left(\frac{\pi n}{4}x\right)dx + \int_2^4 2\cos\left(\frac{\pi n}{4}x\right)dx\right]$$

$$= -\frac{1}{\pi n}\sin\left(\frac{\pi}{2}n\right)$$

or

$$\circledast \quad d_n = \begin{cases} \frac{3}{2} & n = 0 \\ -\frac{1}{\pi n}\sin\left(\frac{\pi}{2}n\right) & n \neq 0 \end{cases}$$

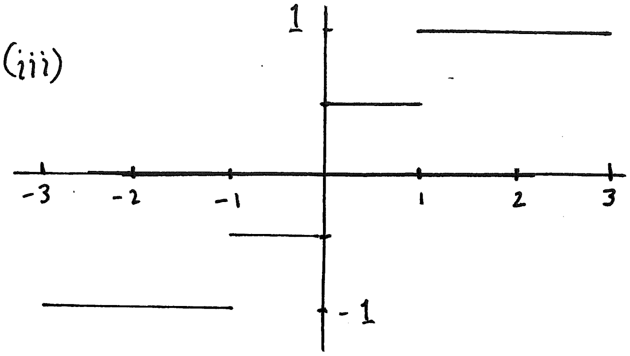
Possible Answers:

1. $f(x) = \frac{3}{2} + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} d_n e^{i\frac{\pi}{4}nx}$

2. $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\frac{\pi}{4}nx}$ (with $\circledast$)

3. $f(x) = \frac{3}{2} + \sum_{n=1}^{\infty} 2d_n \cos\left(\frac{\pi}{4}nx\right)$

(iii)



$\begin{cases} \text{ODD} \\ L=3 \end{cases}$

$d_0 = 0$

$$d_n = \frac{1}{3}\left[\int_0^1 \frac{1}{2}\sin\left(\frac{\pi}{3}nx\right)dx + \int_1^3 \sin\left(\frac{\pi}{3}nx\right)dx\right]$$

$$= \frac{-i}{\pi n}\left[\frac{1}{2}\cos\left(\frac{\pi n}{3}\right) + \frac{1}{2} - (-1)^n\right]$$

$$(\ast) \quad d_n = \begin{cases} 0 & n=0 \\ -\frac{i}{\pi n}\left[\frac{1}{2}\cos\left(\frac{\pi}{3}n\right) + \frac{1}{2} - (-1)^n\right] & n \neq 0 \end{cases}$$

Possible Answers:

1. $f(x) = \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} d_n e^{i\frac{\pi}{3}nx}$

2. $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\frac{\pi}{3}nx}$ (with $(\ast)$)

3. $f(x) = \sum_{n=1}^{\infty} 2i d_n \sin\left(\frac{\pi}{3}nx\right)$

Problem 2 $\quad f(x)$ is Even and $L=1$

$$d_0 = \frac{1}{3}$$

$$d_n = \int_0^1 x^2 \cos(\pi n x)\,dx$$

$=$ (Apply Integration by Parts Twice)

$$= \frac{2}{\pi^2 n^2}(-1)^n$$

$$f(x) = \frac{1}{3} + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{2}{\pi^2 n^2}(-1)^n e^{i\pi n x} = \frac{1}{3} + \sum_{n=1}^{\infty} \frac{4}{\pi^2 n^2}(-1)^n \cos(\pi n x)$$

$$= \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(\pi n x)$$

Extra Credit

Since $\quad x^2 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos(\pi n x) \quad$ for $\quad -1 \leq x \leq 1$

Let $x = 1$:

$$1 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos(\pi n) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}(-1)^n$$

$$= \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\Rightarrow \boxed{\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}}$$

Let $x = 0$:

$$0 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos(0) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

$$\Rightarrow \boxed{\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}}$$